

ONTARIO, OR, USA

WMO: 726837

Lat: 44.021N

Lon: 117.013W

Elev: 2193

StdP: 13.57

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 24162

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.5	(c) 12.9	(d) 0.5	(e) 6.1	(f) 8.5	(g) 6.7	(h) 8.4	(i) 16.2	(j) 24.4	(k) 33.8	(l) 20.1	(m) 34.7	(n) 2.8	(o) 250	(p) 0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 31.1	(c) 98.7	(d) 65.9	(e) 95.8	(f) 64.7	(g) 92.8	(h) 63.9	(i) 68.7	(j) 91.7	(k) 67.0	(l) 90.2	(m) 65.5	(n) 88.6	(o) 4.4	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	87.4	76.4	58.5	79.3	75.1	56.4	73.4	74.2	34.0	92.0	32.6	90.3	31.3	88.2	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.0	(b) 20.1	(c) 17.4	(e) 2.1	(f) 103.8	(g) 8.5	(h) 3.3	(i) -4.1	(j) 106.2	(k) -9.1	(l) 108.1	(m) -13.9	(n) 109.9	(o) -20.0	(p) 112.3		
(a) 24.0	(b) 20.1	(c) 17.4	(e) 1.9	(f) 73.0	(g) 8.7	(h) 2.6	(i) -4.3	(j) 74.9	(k) -9.4	(l) 76.4	(m) -14.3	(n) 77.9	(o) -20.6	(p) 79.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	29.6	35.9	44.8	51.2	60.2	68.7	78.0	75.1	64.7	51.2	39.0	30.1
	DBStd	17.72	8.50	6.40	5.98	6.47	7.52	7.22	5.57	5.26	6.56	6.69	7.38	8.53
	HDD50	2295	633	395	180	63	9	0	0	0	0	63	335	617
	HDD65	5566	1098	814	626	416	185	43	2	4	85	430	781	1082
	CDD50	3198	0	1	19	98	327	561	869	778	442	99	4	0
	CDD65	993	0	0	0	1	38	154	406	316	76	2	0	0
	CDH74	13778	0	0	1	66	673	2110	5466	4145	1254	63	0	0
	CDH80	6993	0	0	0	9	213	955	3112	2201	491	12	0	0
Wind	WSAvg	5.6	4.0	5.4	6.5	7.7	7.0	6.9	5.9	5.1	4.9	5.0	4.5	4.5
Precipitation	PrecAvg	9.8	1.3	0.9	0.9	0.7	0.9	0.7	0.2	0.3	0.4	0.6	1.3	1.5
	PrecMax	16.6	3.7	2.3	3.7	2.9	5.5	2.3	1.2	2.1	1.9	2.1	3.9	4.4
	PrecMin	4.9	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1
	PrecStd	2.5	0.7	0.6	0.7	0.5	1.0	0.6	0.3	0.5	0.4	0.5	0.8	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.4	58.3	70.3	81.0	91.1	99.1	103.6	100.2	93.6	81.2	63.0	52.5
		MCWB	41.4	47.0	51.7	57.7	62.6	66.1	66.8	65.5	63.4	57.9	49.9	45.0
	2%	DB	45.2	53.8	65.4	74.8	85.7	93.1	100.0	97.1	89.8	73.5	57.4	48.1
		MCWB	39.6	44.1	49.5	54.5	59.8	63.6	66.3	65.0	62.1	55.0	47.5	42.1
	5%	DB	42.5	49.8	61.3	70.4	81.4	89.8	97.3	94.2	85.8	69.3	54.0	44.6
		MCWB	37.9	42.1	47.5	52.2	58.5	62.4	65.5	63.9	60.5	52.6	45.3	40.1
	10%	DB	39.1	46.1	57.2	65.7	77.1	85.6	94.2	91.3	81.8	64.6	50.4	40.8
		MCWB	35.6	39.7	45.8	50.0	56.6	60.3	64.6	63.3	59.3	50.6	43.5	37.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.9	48.8	53.5	59.0	64.3	69.2	72.2	69.6	67.4	59.9	52.3	46.7
		MCDB	46.7	56.8	63.7	77.6	85.8	93.9	96.7	91.7	84.7	74.4	59.3	50.5
	2%	WB	40.2	45.0	51.1	55.6	61.8	66.2	69.4	67.6	64.7	56.7	49.3	43.4
		MCDB	44.3	51.8	61.7	71.4	81.2	88.5	92.7	90.6	83.3	70.6	55.9	47.4
	5%	WB	38.2	42.6	49.2	53.3	59.8	64.0	67.8	66.0	62.5	54.1	46.5	40.4
		MCDB	41.8	48.7	59.3	66.4	78.0	85.5	91.6	89.3	80.9	66.6	52.2	44.0
	10%	WB	36.2	40.5	47.0	51.3	57.8	62.1	66.2	64.5	60.3	51.8	44.1	37.3
		MCDB	38.9	45.6	56.5	64.2	74.2	82.2	90.0	87.7	79.1	62.7	49.6	40.5
Mean Daily Temperature Range	5% DB	MDBR	12.3	16.6	22.2	24.8	26.2	27.7	31.1	31.7	30.9	25.6	18.7	12.7
		MCDBR	15.3	21.8	29.0	33.4	33.4	34.3	34.3	35.5	36.4	32.0	23.3	14.9
		MCWBR	10.8	13.6	16.0	16.5	13.7	13.5	13.3	13.5	14.8	16.2	14.5	10.4
	5% WB	MCDBR	13.9	19.1	25.5	29.9	30.1	31.6	32.5	33.3	32.8	28.6	20.9	13.4
		MCWBR	10.1	12.4	14.8	15.2	12.6	13.2	13.5	13.4	13.8	14.9	13.8	9.9
Clear-Sky Solar Irradiance	taub	0.269	0.277	0.292	0.315	0.337	0.335	0.340	0.344	0.321	0.296	0.276	0.269	
	taud	2.562	2.579	2.551	2.467	2.425	2.456	2.448	2.418	2.483	2.555	2.566	2.553	
	Ebn at Noon	280	297	303	302	296	296	293	288	288	284	274	268	
	Edh at Noon	20	24	28	33	36	35	35	35	30	24	20	19	
All-Sky Solar Radiation	RadAvg	479	797	1204	1713	2064	2356	2454	2116	1624	1055	605	393	
	RadStd	75	102	121	134	149	132	89	92	91	84	53	61	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.15	N/A	N/A	+1.67	N/A	N/A	N/A	N/A	+288	+176
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+133	N/A

Nomenclature: See separate page